

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2024 – Present	B.Tech	Indian Institute of Technology Kanpur	7.7/10
2024	Class XII (BIEAP)	Ascent Junior College	96.4%
2022	Class X (CBSE)	Brahmavid The Global School	95.6%

Scholastic Achievements

- Secured the **Merit-cum-Means (MCM) Scholarship** at IIT Kanpur, a merit-based award given to top-performing students across the batch.
- Secured **under 500 rank** in **Convolve 4.0** round 1 among the **8000+** other registered candidates from across the country (2026).

Research Experience

Teleoperation System Research Intern Wadhvani Center for Robotics, IIT Kanpur			Jun’26–Present
Objective	- Develop a low-cost, kinematically equivalent teleoperation leader arm for the AgileX NERO 7-DOF robot, implementing the GELLO framework (Wu et al., RSS 2024) to enable scalable, IK-free imitation-learning data collection. - Deliver a real-time joint-space teleoperation interface with passive gravity feedback near joint limits, avoiding the kinematic singularities of inverse-kinematics interfaces (targeting the published ~100 Hz GELLO baseline).		
Approach	- Derived modified-DH parameters from the manufacturer URDF and scaled link geometry to obtain a kinematically equivalent leader; designed 3D-printable linkages in SOLIDWORKS with passive joint regularization at gravity-dominant joints. - Selected and integrated servos, encoders, and control electronics across all 7 DOF; interfaced the leader to the NERO follower over the vendor’s TCP embedded communication protocol. - Built the real-time interface on the LeRobot framework via a NERO-specific protocol class (nero.py); implemented automated leader–follower zero-offset calibration through known-pose joint detection, and am validating end-to-end teleop for demonstration collection.		
Results	<i>In progress</i>: kinematic-equivalence derivation complete and CAD/electronics integration underway; targeting a first end-to-end teleop demonstration and validated imitation-learning data collection on the NERO arm by expected completion (Aug’26).		

Work Experience

Full Stack Developer Intern Apni Vidya			May’26–Present
Objective	Designed and built a multi-role web platform to automate coaching institute workflows, centralising student enrollment, attendance tracking, and fee management.		
Execution	- Engineered a Node.js/Express REST API (78 endpoints) backed by a 24-table PostgreSQL schema and a React SPA, implementing JWT authentication and RBAC. - Built a QR-driven enrollment pipeline and an MCQ engine featuring server-side auto-grading, dynamic submission re-ranking, and configurable per-topic accuracy analysis. - Integrated Razorpay with HMAC-SHA256 verification, utilizing SELECT FOR UPDATE row locks to guarantee idempotent fee reconciliation and eliminate race conditions. - Architected a pluggable MSG91/Gupshup notification layer for scheduled reporting, and enforced security via bcrypt, API rate limiting, and environment-configurable CORS.		
Impact	Streamlined end-to-end institute operations, delivering a highly concurrent platform capable of processing payments and scheduled tasks without manual oversight.		

Key Projects

Signals & Systems Course Project Course Project, EE200 Prof. Tushar Sandhan, IIT Kanpur			May’26–June’26
Objective	End-to-end signal-processing project covering 2D Fourier image restoration, biomedical ECG arrhythmia detection, and an audio fingerprinting system with a live deployed app.		
Execution	Q1 (Image): Recovered a hidden message from a periodically-corrupted image using 2D DFT, frequency-domain notch filtering, and inverse DFT; applied Sobel convolution for edge detection. Q2 (ECG): Detected arrhythmia onset in a 20-s Holter recording via normalized cross-correlation with a 200-sample healthy-beat template; verified with spectrogram and RR-interval analysis. Q3 (Audio): Built full pipeline in Python STFT spectrogram, local-maxima peak detection, combinatorial $(f_1, f_2, \Delta t)$ hashing; indexed 5.25M hashes across 50 songs; deployed an interactive Streamlit app (ee200-shazam-harsh.streamlit.app).		
Impact	Built and deployed 3 working signal-processing systems across image, ECG, and audio; the live audio identifier recognizes songs in ~75 ms with a >1000× match margin and stays correct down to SNR = −5 dB.		

Objective	To design and build a 6-DOF robotic arm controlled via real-time hand gestures using computer vision, with a focus on contactless, intuitive operation.
Execution	- Designed a 21-part mechanical assembly using 3D-printed PLA components, ACP base plates, and a metal shaft; performed detailed torque calculations for all 6 joints. - Implemented real-time gesture recognition using Python + OpenCV + MediaPipe to detect 21 hand landmarks and map positions to servo angles.
Impact	Achieved real-time contactless arm control with a payload capacity of ~120-150g, constrained by shoulder joint torque. Total prototype value estimated at Rs 6,155 with an out-of-pocket cost of only Rs 2,250.

Objective	To build a robust time series forecasting pipeline using machine learning, deep learning, and game-theoretic ensemble methods for financial and temporal data.
Execution	- Analyzed temporal datasets extracting trend, seasonality components; engineered lag features and rolling statistics using NumPy, Pandas, and Matplotlib. - Trained ML and deep learning models (Random Forest, LSTM/GRU) and implemented a multi-agent weighted ensemble for combining predictions across agents.
Impact	Competed in a Kaggle-style forecasting competition, building a complete pipeline from data preprocessing to final ensemble prediction.

Objective	To develop practical DBMS skills covering relational design, SQL, and NoSQL through a structured 8-week mentorship program.
Execution	- Designed normalized relational schemas using ER diagrams, keys, and constraints in PostgreSQL and draw.io. - Wrote advanced SQL queries involving joins, aggregates, window functions, CTEs, and transactions with ACID properties. - Studied CAP theorem and SQL vs NoSQL trade-offs to inform database selection for real-world use cases.
Impact	Cleared Midterm and End-Term evaluations covering schema design, query optimization, and system trade-off analysis.

- Developed a C++ **CLI-based** Car Rental System using **OOP** to enable multi-role access and persistent data handling.
- Employed the **object-oriented programming** principles of Polymorphism, Inheritance, Encapsulation and Abstraction.
- Maintained modular databases using **file handling in C++** and rigorously tested the software on edge cases for robustness.

- Fabricated the **palm and arm assembly** from 0.5mm mild steel sheet with precise bending and joining to achieve the tapered geometry (200×180mm base, 150mm top).
- Assembled **4 articulated fingers and thumb** connected via 3mm mild steel rods, nuts, bolts, and washers enabling a tendon-driven **finger contraction mechanism**.
- Contributed to full **gauntlet assembly** integrating 7 distinct components across a 6-member team through sheet metal forming and turning processes.

Technical Skills

Languages & ML/Data	C C++ Python MATLAB SQL NumPy Pandas Matplotlib scikit-learn TensorFlow
CAD, Simulation & Robotics	SOLIDWORKS Fusion 360 AutoCAD MATLAB Simulink OpenCV MediaPipe
Databases & Dev Tools	PostgreSQL GitHub $\LaTeX$ VSCode Jupyter Notebook Streamlit Arduino IDE

Relevant Courses

Fundamentals of Computing	Introduction to Electronics	Signals Systems and Networks
Fluid Mechanics and Rate Process	Mechanics of Solids	Manufacturing Process
Introduction to Aerospace Engineering	Incompressible Aerodynamics	Dynamics
Engineering Graphics	Oscillations And Waves	Experiments in Aerospace Engineering
Single Variable Calculus	Application of Single Variable Calculus and Several Variable Calculus	Linear Algebra
Ordinary Differential Equations	Partial Differential Equations	Complex Variables
Introduction to Linguistics	Introduction to Management	English Language and Communication

Positions of Responsibility

- Appointed **Associate Industry Connect Head** at **SAE**, the official body of the **Department of Aerospace Engineering, IIT Kanpur**, leading industry-academia engagement alongside the Industry Connect Head.
- Responsible for liaising with **aerospace and defence companies** to organise **industrial visits, technical talks, sponsored workshops**, and collaborative project opportunities for the UG and PG aerospace community.

- Coordinated participant onboarding and end-to-end event execution for **IARC** at Techkriti'26, managing **65+ participants** across maze-solving and precision navigation challenges.
- Managed on-ground operations and logistics for **Atlassian × Techkriti FastTrack Hackathon** at Techkriti'26, overseeing smooth execution for **50+ participants** competing for a Rs 2,00,000 prize pool.

Extra-Curricular Activities

- Awarded **Best Bowler** at ICC 2025 (Institute Cricket Championship), IIT Kanpur, finishing as the highest wicket-taker in the tournament.
- Completed Prarambhik Part-1 & Part-2 in Fine Arts (Painting) from **Pracheen Kala Kendra, Chandigarh** with **First Division and Distinction** in Theory & Practical (2018-19, 2019-20).
- Clinched **Gold Medal** in Mime at **Galaxy'25**, Annual Cultural Festival, IIT Kanpur.
- Secured **Silver Medal** in Nukkad Natak at **Galaxy'25**, Annual Cultural Festival, IIT Kanpur.
- Secured **Bronze Medal** in Cricket at **Inferno'25** Inter-Hall Sports Tournament, IIT Kanpur.